Experiment 12.1.1

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Algorithm:

1. Start
2. Input an integer n (number of operations)
3. Initialize an empty dictionary
   contacts ← { }

4. Repeat the following steps for i = 1 to n:

4.1 Read the operation (split into parts)

4.2 If operation is "ADD":

- Extract name and phone_number
- If name already exists in contacts:
  - Update the phone number
- Else:
  - Add new entry
- contacts[name] ← phone_number

4.3 Else if operation is "REMOVE":

- Extract name
- If name exists in contacts:
  - Remove it
- Else:
  - Ignore operation

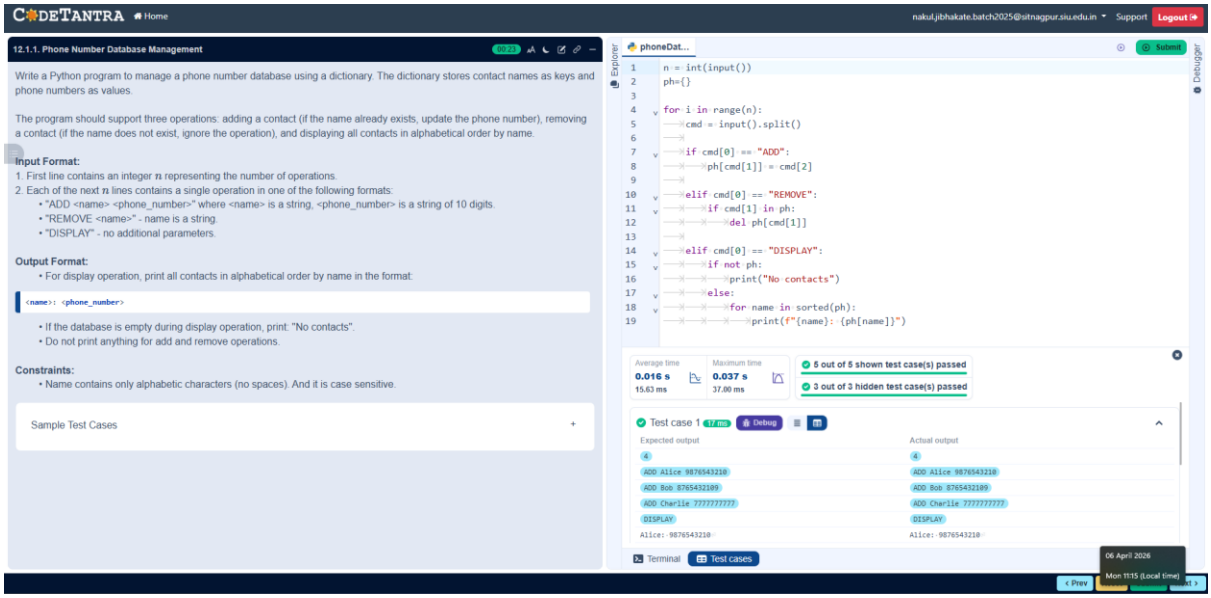
4.4 Else if operation is "DISPLAY":

- If contacts is empty:
  - Print "No contacts"

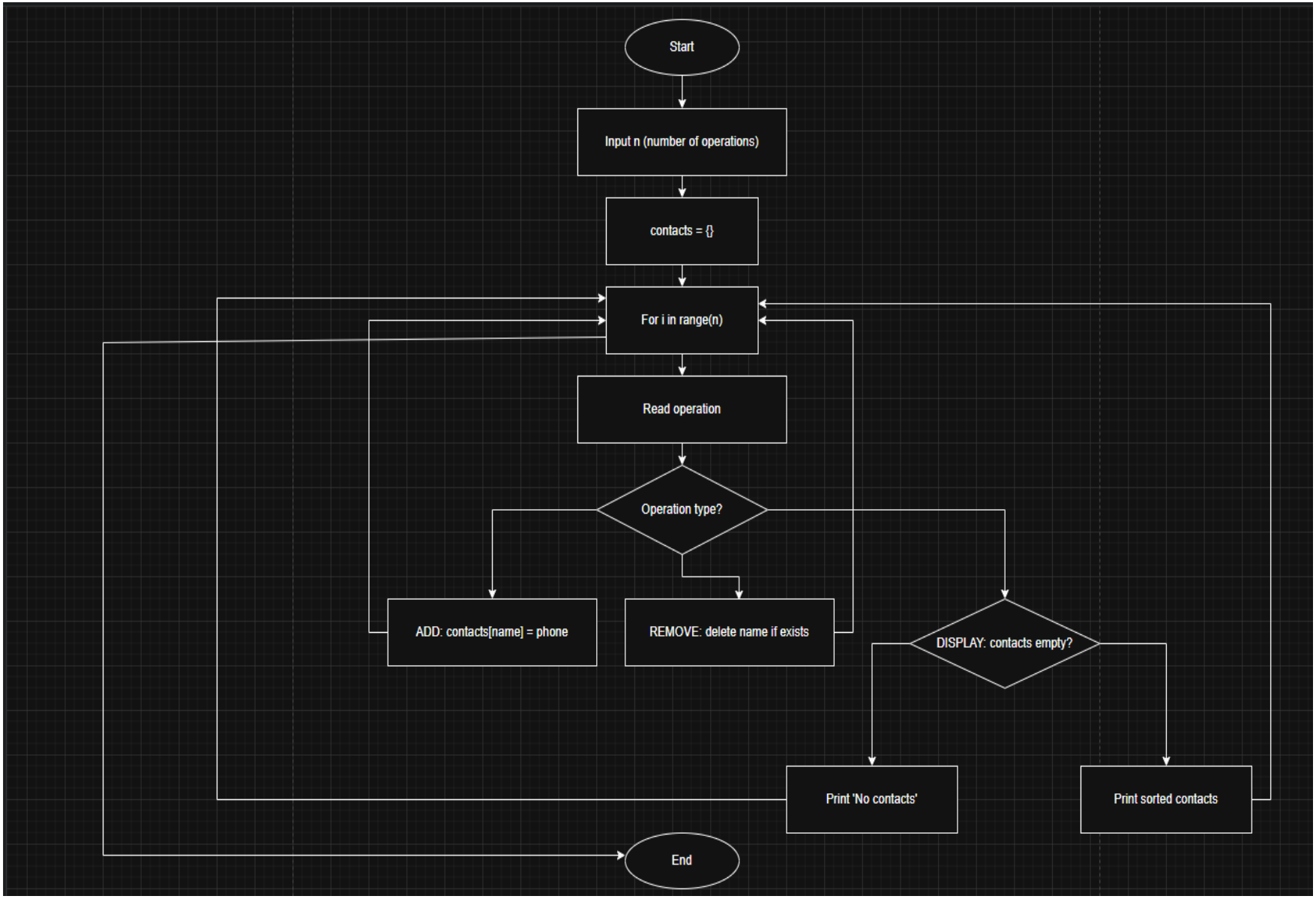
- Else:
    - Sort all names in alphabetical order
    - For each name in sorted list:
        - Print name : phone_number

5. End loop
6. End

Execution:



Flowchart:


Start
Input n (number of operations)
contacts = {}
For i in range(n)
Read operation
Operation type?
ADD: contacts[name] = phone
REMOVE: delete name if exists
DISPLAY: contacts empty?
Print 'No contacts'
Print sorted contacts
End